

1. Difference between Array and Linked List:

Basis	Array	Linked List
Memory allocation	Contiguous (continuous memory)	Non-contiguous (scattered memory)
Size	Fixed size	Dynamic size
Access time	Fast (direct access using index)	Slow (sequential access)
Insertion / Deletion	Difficult (requires shifting elements)	Easy (just change links)
Memory usage	Less memory (no extra pointers)	More memory (extra pointer for link)
Cache friendliness	High	Low
Example	<code>int a[10];</code>	Nodes connected using pointers

2. Difference between Stack and Queue

Parameter	Stack	Queue
Principle	Last-In, First-Out (LIFO)	First-In, First-Out (FIFO)
Insertion	Called push; occurs at the top.	Called enqueue; occurs at the rear.

Deletion	Called pop; occurs from the top.	Called dequeue; occurs from the front.
Pointers	Uses a single pointer (to the <i>top</i>).	Uses two pointers (<i>front</i> and <i>rear</i>).
Real-world Analogy	A stack of plates.	People in a line/queue at a counter.
Use Cases	Function call management, "undo" features, expression evaluation.	Task scheduling, print job management, breadth-first search algorithms.

3. Difference between Top-Down Approach and Bottom-Up Approach

Basis	Top-Down Approach	Bottom-Up Approach
Direction	From main problem to sub-problems	From small sub-problems to main problem
Design method	Stepwise refinement	Composition of modules
Focus	Overall structure first	Detailed components first

Implementat ion	Easier to understand	Easier to implement
Used in	Recursive solutions, algorithm design	Dynamic programming, system design
Testing	Requires stubs	Requires drivers

4. Difference between Big-O (O), Big-Omega (Ω), and Big-Theta (Θ)

Aspect	Big-O (O)	Big-Omega (Ω)	Big-Theta (Θ)
Type of bound	Upper bound	Lower bound	Tight bound
Case	Worst case	Best case	Average / exact
Performance view	Maximum time	Minimum time	Exact growth
Usage	Performance guarantee	Minimum cost	Precise analysis

Notation	Meaning	Represents
Big-O (O)	Upper bound	Worst-case time complexity
Big-Omega (Ω)	Lower bound	Best-case time complexity
Big-Theta (Θ)	Tight bound	Average / exact time complexity